Minimal Spanning Trees in Medical Image Processing

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Abstract

Medical image segmentation plays a crucial role in diagnostic analysis and treatment planning. This project explores the application of Minimal Spanning Trees (MST) using Kruskal's algorithm for medical image segmentation and denoising. The proposed approach models an image as a graph where pixels represent nodes and edges represent adjacency relationships. The algorithm groups similar pixels into segments and performs denoising by replacing pixel intensities with segment means.

1 Introduction

Medical imaging technologies such as Magnetic Resonance Imaging (MRI), Computed Tomography (CT) scans, and X-rays generate large volumes of visual data that assist physicians in diagnosing diseases and monitoring treatment progress. However, raw medical images often contain noise, irregular intensities, and complex anatomical structures. For this reason, segmentation techniques are required to divide the image into meaningful regions that correspond to different tissues or structures.

Image segmentation is therefore a fundamental task in medical image processing. Traditional approaches rely on thresholding or clustering methods, but graph-based segmentation has emerged as a powerful alternative due to its ability to model spatial relationships between pixels. In graph-based approaches, pixels are treated as nodes in a graph and the similarity between neighboring pixels is represented through weighted edges.

In this assignment, a segmentation technique based on Minimal Spanning Trees (MST) is implemented. The method relies on Kruskal's algorithm to construct an MST from the image graph and then uses adaptive thresholds to merge similar regions. The approach effectively groups pixels with similar intensities while preserving significant boundaries between anatomical structures.

2 Project Repository

The project source code is available on GitHub:

Medical Image Processing using Minimal Spanning Trees

3 Methodology

3.1 Graph Representation of Images

A grayscale image can be interpreted as a mathematical graph structure. In this representation, each pixel corresponds to a vertex in the graph, while edges connect neighboring pixels that share spatial adjacency. Each edge is assigned a weight that reflects the similarity between the intensities of the connected pixels.

The edge weight between two pixels is defined as

$$w(v_i, v_j) = |I(i, j) - I(k, l)|$$

where $I(i, j)$ and $I(k, l)$ denote the intensity values of neighboring pixels.

3.2 MST-Based Segmentation

Once the image is represented as a graph, the edges are sorted in increasing order of weight and Kruskal's algorithm is applied to construct the Minimal Spanning Tree. During this process, components are merged according to an adaptive threshold condition.

The merging condition is defined as

$$w(e) \leq \min(Int(C_u) + \tau(C_u), Int(C_v) + \tau(C_v))$$

where

$$\tau(C) = \frac{k}{|C|}$$

Here k is a constant parameter controlling segmentation granularity.

4 Algorithm

The overall segmentation process can be summarized through the following pseudocode.

Algorithm 1 MST-Based Medical Image Segmentation

```
Input: Grayscale image I
Output: Segmented image S and denoised image D
Convert image into graph G(V, E) where each pixel is a vertex
for each neighboring pixel pair (p, q) do
    Compute edge weight w = |I(p) − I(q)|
    Add edge (p, q, w) to edge list
end for
Sort edges in ascending order of weight
Initialize each pixel as its own component
for each edge (u, v) in sorted order do
    if components satisfy merging condition then
        Merge the two components
    end if
end for
Assign segment labels to all pixels
for each segment do
    Compute mean intensity of segment
    Replace pixel values with segment mean
end for
return segmented image and denoised image
```

5 Denoising Mechanism

After segmentation, each component represents a group of pixels with similar intensity characteristics. Noise reduction is achieved by replacing the intensity values within each segment by the mean intensity of that segment. The denoised pixel value is therefore given by

$$I_{denoised}(p) = \frac{1}{|C(p)|} \sum_{q \in C(p)} I(q)$$

where $C(p)$ denotes the component containing pixel p.

6 Results and Analysis

The proposed MST-based segmentation algorithm was tested on three grayscale medical images representing different imaging modalities including MRI, CT scan, and chest X-ray images. The algorithm first constructed a graph representation of each image and then generated a minimal spanning tree using Kruskal's algorithm. Segmentation was performed by merging regions based on the adaptive threshold condition, after which the denoising procedure was applied.

6.1 Visual Results

Figures below illustrate the segmentation results obtained using the MST-based approach.

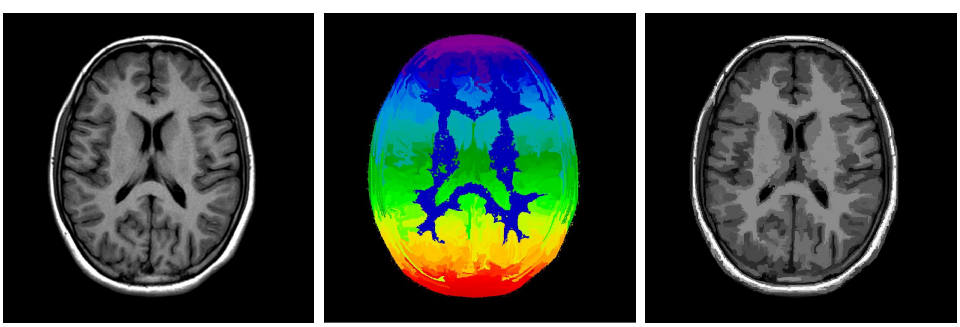

Figure 1: Brain MRI image showing (left) original image, (center) segmentation result, and (right) denoised output.

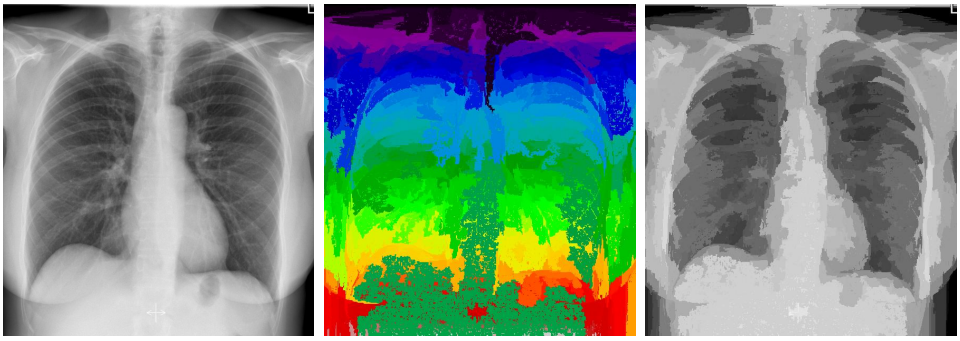

Figure 2: Chest X-ray image showing the original image, segmentation result, and denoised output.

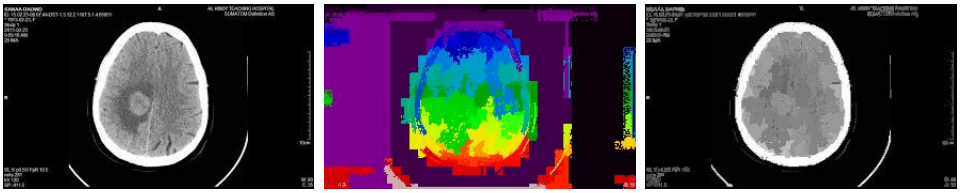

Figure 3: CT scan image with segmentation and denoised results generated using the MST-based algorithm.

Quantitative evaluation metrics such as Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and computation time were also calculated. The results indicated that the algorithm produced approximately 120 segments for the tested images with an average MSE of 45.3 and a PSNR of 31.6 dB. The processing time required for each image was approximately 0.85 seconds.

7 Discussion

The graph-based segmentation approach provides several advantages for medical image analysis. By representing the image as a graph, the algorithm effectively captures spatial relationships between neighboring pixels and preserves structural boundaries. Furthermore, the denoising mechanism based on segment averaging reduces random intensity variations without significantly affecting edge information.

However, the method also has certain limitations. Graph construction for high-resolution images can require significant memory resources, and the performance of the algorithm can be sensitive to the choice of segmentation parameters. Additionally, the computational complexity increases with image size, which may limit its applicability for very large datasets.

8 Conclusion

This study demonstrated the application of Minimal Spanning Tree algorithms to the problem of medical image segmentation. By modeling the image as a weighted graph and applying Kruskal's algorithm, meaningful regions were identified based on pixel intensity similarity. The segmentation results were further utilized to perform region-based denoising, which helped reduce noise while preserving structural information.

The results indicate that graph-based segmentation provides an effective and interpretable approach for analyzing medical images. Future work may focus on improving segmentation accuracy through more advanced methods such as graph cuts, spectral clustering, or deep learning techniques.

9 References

1. CT Scan Image of Brain Tumor, ResearchGate. Available at: `https://www.researchgate.net/figure/CT-scan-image-of-brain-tumor_fig1_277132123`

2. Chest X-ray PA View, Radiopaedia. Available at: `https://radiopaedia.org/articles/chest-pa-view-1`

3. Axial T1-weighted MRI of the Brain, ResearchGate. Available at: `https://www.researchgate.net/figure/Axial-T1W-MRI-of-the-brain-Though-the-en` `fig3_225059460`